

AdaViP: Task-Conditioned Visuotactile Perception via Hypernetworks for Contact-Rich Manipulation

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Abstract—Contact-rich manipulation requires a robot to change what it perceives and trusts as an interaction evolves. This paper studies task-conditioned adaptive visuotactile perception for real-world robotic manipulation.

I. INTRODUCTION

Contact-rich manipulation underpins many practical robotic skills, including insertion, peeling, wiping, tool use, and dexterous in-hand manipulation. These tasks cannot be solved reliably from vision alone: visual observations provide global geometry and free-space context, while tactile sensing reveals local contact, deformation, friction, and slip that may be occluded or too subtle to infer from images [1]–[4]. Proprioception further anchors these signals in the robot’s motion and force state. Effective manipulation therefore depends on combining heterogeneous sensory evidence throughout an interaction.

The central difficulty is that the value of each modality changes over time. During approach, visual geometry and object pose dominate. Near contact onset, small tactile transients and changes in end-effector state become decisive. During sustained contact, the policy must jointly interpret global visual context, local tactile deformation, and proprioceptive feedback. A perception stack with fixed parameters must nevertheless apply the same feature extraction strategy across all of these phases. This creates a structural mismatch between a changing physical interaction and a static sensory representation.

Existing visuotactile policies alleviate this mismatch mainly after features have already been computed. Concatenation and cross-attention combine modality embeddings; tactile gates suppress irrelevant touch; slow-fast controllers introduce high-rate tactile correction; and policy-consensus methods route among modality-specific action predictions [5]–[9]. Larger tactile-language and world-model systems similarly introduce contact-aware tokens, modality selection, or predictive objectives [10]–[16]. These mechanisms improve fusion and control, but they typically leave the upstream perception mapping fixed after training. As a result, the model can change how it weights a visual or tactile feature without changing what that feature extractor is optimized to represent at the current task stage.

Our core insight is that adaptive multimodal control should begin before policy inference. Rather than only deciding which modality to trust, a robot should also adjust what each modality extracts according to the task and the evolving interaction state. Conditional computation provides a suitable mechanism: hypernetworks can generate context-dependent parameters for

another network [17], while related approaches such as dynamic filters and FiLM demonstrate that feature computation can be conditioned without duplicating a complete policy [18], [19]. This suggests a compact alternative to maintaining separate perception-policy stacks for different contact regimes.

We introduce AdaViP, a task-conditioned visuotactile perception framework for contact-rich manipulation. AdaViP uses hypernetworks to adapt the multimodal perception pathway from task and execution context before the resulting representation is passed to a shared action policy. The formulation is intentionally centered on adaptive perception rather than policy selection: task conditioning changes the representation delivered by the visual, tactile, and proprioceptive streams, while the downstream policy remains shared across the rollout. This design targets heterogeneous sensory observations whose relevance changes sharply around contact, while avoiding a separate specialist policy for every task phase.

We evaluate AdaViP entirely on real robots, where sensor noise, occlusion, contact uncertainty, and embodiment-specific dynamics are present by construction. The primary experiment compares AdaViP with static visuotactile perception and adaptive-fusion baselines across multiple contact-rich tasks. A controlled ablation then isolates the roles of task conditioning, execution context, and hypernetwork-based adaptation. Following recent efforts toward systematic visuotactile evaluation [20], this protocol is designed to test both the overall manipulation benefit and the specific claim that adapting perception itself provides value beyond static feature extraction followed by dynamic fusion.

The contributions of this work are:

- We formulate contact-rich manipulation as a problem of task-conditioned visuotactile perception, in which the sensory representation should change with the task and interaction state.
- We propose AdaViP, a hypernetwork-based framework that adapts heterogeneous visual, tactile, and proprioceptive perception before action prediction while retaining a shared policy backbone.
- We design a real-robot evaluation with a main comparison and targeted ablations to distinguish adaptive perception from static perception and downstream adaptive fusion.

II. RELATED WORK

A. Tactile Sensing and Representation Learning

High-resolution optical tactile sensors convert deformation of a compliant surface into image-like observations. GelSight

established a widely used design for estimating local geometry and force [2]; DIGIT reduced the cost and form factor for in-hand manipulation [4]; GelSlim 3.0 jointly measures shape, force, and slip in a compact finger [21]; and Soft-Bubble grippers combine dense visuotactile geometry with mechanical compliance [22]. Such sensors have enabled visuotactile tool use [23], tactile state estimation [24], and multimodal object tracking under occlusion [25].

A parallel line of work studies representations that are less tied to a single task or sensor. Self-supervised objectives align vision and touch for contact-rich manipulation [26], while contrastive pretraining improves multimodal transfer [27]. Sparsh learns general-purpose tactile encoders across several vision-based sensors [28], and AnyTouch unifies static and dynamic signals across multiple optical tactile platforms [29]. Other approaches explicitly model contact dynamics or physical state, including TactileVAD [30], Robot Synesthesia [31], and tactile pretraining for cable insertion [32]. These methods improve the information available to a policy; AdaViP instead studies how the mapping from sensory input to policy representation should change with task context and execution phase.

B. Visuotactile Policy Learning

Visuotactile policy learning spans grasp prediction, local feedback control, dexterous manipulation, and long-horizon imitation. Touch improves grasp-outcome prediction when combined with vision [3], supports local cloth control through visuotactile affordances [33], and enables in-hand manipulation when visual state estimates are ambiguous [34]. For dexterous imitation, Learning Visuotactile Skills with Two Multifingered Hands collects demonstrations with tactile-equipped hands [35], while 3D-ViTac fuses visual and tactile observations in a common 3D representation [36].

Sequence-policy backbones have accelerated this direction. Diffusion Policy models multimodal action distributions through conditional denoising [37], and ACT predicts temporally coherent action chunks [38]. RDP augments slow action generation with a fast tactile feedback path [5]; ReTac-ACT introduces cross-modal attention and state-conditioned tactile gating [6]; Seeing Touch from Motion models tactile temporal change and cross-modal interaction [7]; and Multi-Modal Policy Consensus routes among modality-specific diffusion policies [8]. GelFusion uses vision-led cross-attention to reduce visual degradation during fusion [9]. These methods primarily improve policy reactivity, tactile encoding, or downstream fusion. AdaViP addresses the complementary question of adapting the perception transformation that precedes the shared policy.

The same issue is emerging in larger multimodal policies. TLA and VTLa ground tactile sequences with language for contact-intensive insertion [10], [11]; FuSe adapts generalist robot policies to heterogeneous sensors through language grounding [12]; and TacVLA activates tactile tokens through contact-aware gating [13]. T-Rex scales tactile-reactive manipulation through variable-rate tactile processing [14], whereas

OmniVTA and Dream-Tac introduce predictive visuotactile world or world-action models [15], [16]. These systems reinforce the need for selective multimodal computation, but their scope differs from explicitly generating task- and phase-dependent perception transformations for a compact imitation-learning policy.

C. Task-Conditioned and Adaptive Perception

Conditional computation changes a model’s parameters or activations as a function of context. Hypernetworks generate weights for a target network [17]; Dynamic Filter Networks generate sample-specific convolutional filters [18]; FiLM modulates intermediate features with conditional affine transformations [19]; and Gated Multimodal Units learn multiplicative gates for information fusion [39]. These mechanisms provide different trade-offs between expressivity, computational cost, and architectural disruption.

In manipulation, task information is often concatenated with a task-agnostic observation representation or introduced only at the policy level. HyperTASR instead conditions visual representation transformations on both task specification and progression state [40]. AdaViP follows the broader premise that perception should adapt during execution, but addresses a different technical setting: heterogeneous vision, touch, and proprioception streams whose informativeness changes around contact. The goal is therefore not merely task-aware visual selection or scalar modality weighting, but context-dependent reconfiguration of the multimodal representation supplied to a shared control policy.

III. METHOD

A. Method Overview

Given synchronized visual observation o_v , tactile observation o_t , and proprioceptive state o_p , AdaViP first extracts base representations z_v^0 , z_t^0 , and z_p^0 with frozen modality-specific encoders. A set of conditioned HyperNet modules, shown compactly as one generator block in Fig. 1, receives the task embedding together with an execution-progress context derived from the current base representations. The HyperNets generate only the parameters of three modality-specific transformation encoders, while the corresponding decoders remain ordinary shared learned modules. This produces task- and phase-adaptive representations z_v , z_t , and z_p . A cross-modal transformer exchanges information among the transformed streams, after which a fourth generated fusion encoder and a shared fusion decoder form z_{fused} . The resulting representation is passed to an unchanged policy backbone, so AdaViP changes the upstream perception mapping rather than replacing or specializing the action policy.

IV. EXPERIMENTS

A. Experimental Setup

All experiments are conducted on a physical single-arm manipulation platform equipped with a parallel-jaw gripper. The sensing suite consists of one wrist-mounted RGB camera for local tool-object geometry, one fixed third-person RGB

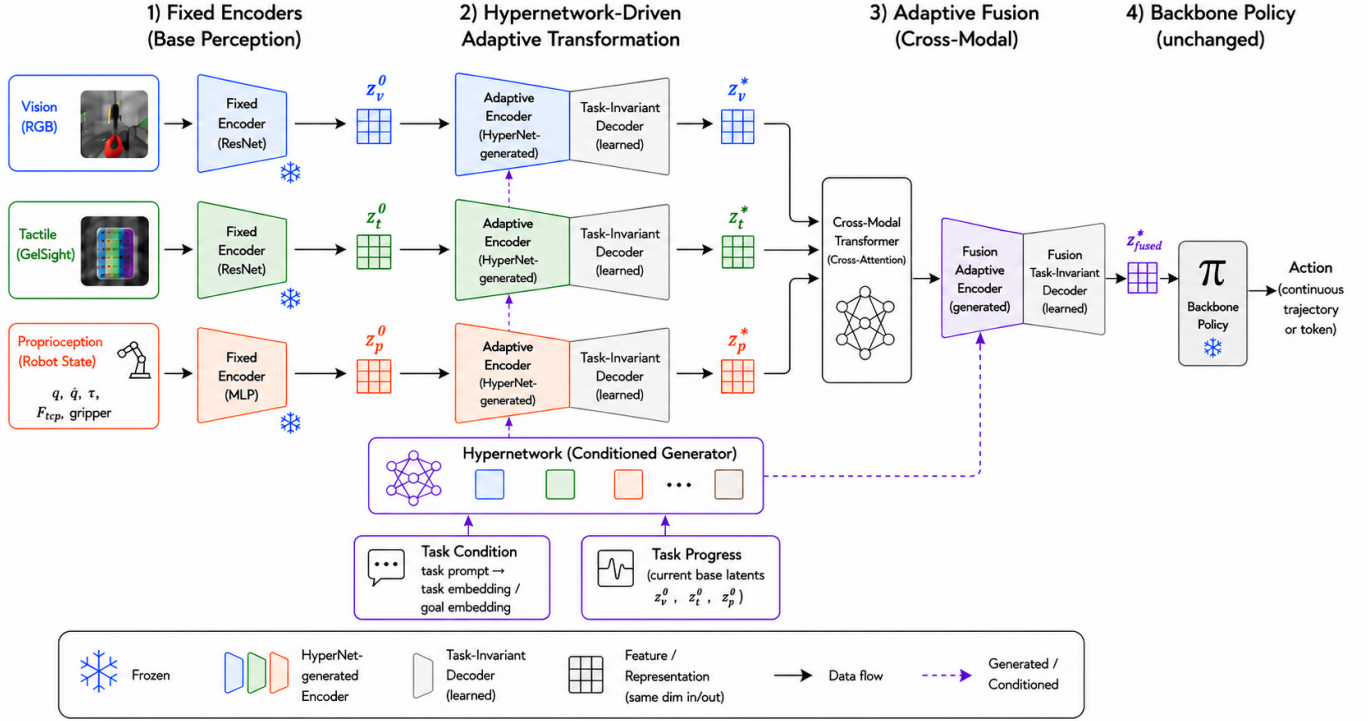


Fig. 1. Overview of AdaViP. Frozen modality encoders produce base visual, tactile, and proprioceptive representations. Task and execution context condition HyperNet modules that generate the transformation encoders for each modality and for fusion; the paired decoders remain shared and task-invariant. Cross-modal attention then integrates the transformed streams before an unchanged policy backbone predicts the action.

camera for global workspace context, GelSight optical tactile sensing at the gripper, and robot proprioception including end-effector pose and gripper state. Depth may be recorded for future analysis, but all methods in the primary comparison receive the same RGB, tactile, and proprioceptive observations. The exact robot, camera, and GelSight models will be specified after the final hardware configuration is confirmed.

We study five tasks with distinct contact patterns: Peeling, which requires precise contact establishment and disturbance recovery; Wiping, which requires sustained surface contact; Occluded Object Picking, in which visual evidence degrades after contact; Threaded Assembly, performed with a large 3D-printed screw-and-nut pair and requiring alignment, axial contact, and rotation; and Puzzle Insertion, which requires contact-guided correction under tight geometric constraints. Demonstrations are collected by teleoperation and shared across all methods.

For a controlled comparison, VT-DP, RDP, Policy Consensus, ReTac-ACT, and the AdaViP variants are trained and evaluated using the same robot embodiment, sensor streams, demonstrations, action representation, train-validation split, and rollout protocol. Methods originally developed with different tactile sensors or multi-arm systems are adapted to this unified single-arm GelSight setup while retaining their defining algorithmic components. The primary metric is task success rate, reported per task and averaged across the five tasks. The

number of demonstrations, training seeds, and evaluation trials will be fixed before data collection and reported in the final manuscript.

B. Main Results

Table I compares complete visuotactile policies under a unified real-robot setup. All methods use the same demonstrations, camera and GelSight observations, action space, and rollout protocol. AdaViP is evaluated as a plug-and-play perception module on both VT-DP and RDP backbones.

C. Ablation and Design Analysis

Tables II and III separate two questions. Table II compares the parameterization of the adaptive transformation while keeping the HyperNet input, cross-attention fusion, policy backbone, and training protocol fixed. Static AE uses the same encoder-decoder without dynamic parameter generation; FiLM predicts feature-wise affine modulation; HyperLoRA predicts low-rank weight updates; and AdaViP predicts the transformation-encoder parameters. Table III ablates cross-attention fusion and the information available to the HyperNet. In all conditioning ablations, the downstream policy still receives the complete visual, tactile, and proprioceptive observation; only the HyperNet input changes.

TABLE I

MAIN REAL-ROBOT RESULTS. SUCCESS RATE (%) IS EVALUATED ON FIVE CONTACT-RICH TASKS USING THE SAME ROBOT, SENSOR SETUP, DEMONSTRATIONS, AND TRIAL PROTOCOL FOR ALL METHODS. THREADED ASSEMBLY USES A LARGE 3D-PRINTED SCREW-AND-NUT PAIR.

Method	Peeling	Wiping	Occluded Picking	Threaded Assembly	Puzzle Insertion	Avg.
VT-DP	—	—	—	—	—	—
RDP [5]	—	—	—	—	—	—
Policy Consensus [8]	—	—	—	—	—	—
VT-DP + AdaViP	—	—	—	—	—	—
RDP + AdaViP	—	—	—	—	—	—

TABLE II

ADAPTIVE TRANSFORMATION PARAMETERIZATION ON THE VT-DP BACKBONE.

Variant	Avg. SR (%)
Static AE	—
FiLM	—
HyperLoRA	—
AdaViP	—

TABLE III

ARCHITECTURE AND HYPERNET-CONDITIONING ABLATIONS ON THE VT-DP BACKBONE.

Variant	Avg. SR (%)
w/o Cross-Attention Fusion	—
Vision-Proprio Conditioning	—
Tactile-Proprio Conditioning	—
Oracle Progress Conditioning	—
AdaViP (Full)	—

V. CONCLUSION

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